

Problem-Solving and Programming Completed SPARK Reference

HarbourView Community Library - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: HarbourView Community Library

Sector: Education

Questions being answered

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the overdue amount for each library member and the total fines from all records processed.

How SPARK completed it

- State what information the program receives.
- State the main calculation or decision the program performs.
- State the result the program must display.

Completed result

Problem statement: Calculate the overdue amount for each library member and the total fines from all records processed.

2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

How SPARK completed it

- List inputs, processes and outputs before writing pseudocode.
- Use indentation to show the selection and loop structure.
- Keep variable names consistent from the algorithm through to the program.

Completed result

Inputs: Member name, Days late, Member type

Processes: Calculate base fine at JMD 100 per day, Apply 25% student concession, Accumulate total fines

Outputs: Member name, Days late, Fine, Total fines

Pseudocode:

START

SET totalFines <- 0

INPUT recordCount

FOR record <- 1 TO recordCount

INPUT memberName, daysLate, memberType

SET fine <- daysLate * 100

IF memberType = 'STUDENT' THEN

SET fine <- fine * 0.75

ENDIF

SET totalFines <- totalFines + fine

DISPLAY memberName, daysLate, fine

NEXT record

DISPLAY totalFines

STOP

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

How SPARK completed it

- Choose values that exercise each branch of the selection statement.
- Follow the pseudocode in order and record important variable values.
- Calculate the expected output before running the program.

Completed result

The completed SPARK trace table includes the project's supplied normal examples and adds a boundary/validation discussion in the completed PDF.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as LibraryFines_YourFullName.pas.

How SPARK completed it

- Create variables using suitable data types.
- Translate each pseudocode step into program statements in the same logical order.
- Compile or run often so errors are found in small sections.
- Use the same test data from the trace table.

Completed result

The SPARK reference implementation is LibraryFines_YourFullName.pas. The completed PDF includes the full source code.

2. Test the program and capture evidence that the expected and actual results agree.

How SPARK completed it

- Run normal test data first.
- Run the boundary case.
- Run an invalid or unusual input when the program includes validation.
- Capture screenshots showing both the entered data and the program result.

Completed result

The completed SPARK reference compares expected trace-table outputs with the program output and records the result of each test.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

How SPARK completed it

- Create a cover page and table of contents.
- Keep the section order the same as the task sequence.
- Use readable code formatting and captions for screenshots.
- Export the final document to PDF and check every page.

Completed result

Completed documentation filename: ProgramDocumentation_YourFullName.pdf. The SPARK section PDF demonstrates the required order and evidence.

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Problem definition

Calculate the overdue amount for each library member and the total fines from all records processed.

Input - Process - Output

Inputs	Processes	Outputs
Member name Days late Member type	Calculate base fine at JMD 100 per day Apply 25% student concession Accumulate total fines	Member name Days late Fine Total fines

Pseudocode

```
START
SET totalFines <- 0
INPUT recordCount
FOR record <- 1 TO recordCount
  INPUT memberName, daysLate, memberType
  SET fine <- daysLate * 100
  IF memberType = 'STUDENT' THEN
    SET fine <- fine * 0.75
  ENDIF
  SET totalFines <- totalFines + fine
  DISPLAY memberName, daysLate, fine
NEXT record
DISPLAY totalFines
STOP
```

Trace table and test data

Name	Days Late	Member Type	Fine
Janelle Rose	3	STUDENT	225
Paul Green	5	ADULT	500
Kemar White	0	STUDENT	0

Testing note

The normal data above should be supplemented with a boundary case and an invalid or unusual case where the program includes validation. Expected output should be calculated before the program is run.

Pascal reference implementation

```
program LibraryFines;
var
  i, recordCount, daysLate: integer;
  memberName, memberType: string;
  fine, totalFines: real;
begin
  totalFines := 0;
  write('Number of records: '); readln(recordCount);
  for i := 1 to recordCount do
    begin
      write('Member name: '); readln(memberName);
      write('Days late: '); readln(daysLate);
      write('Member type: '); readln(memberType);
      fine := daysLate * 100;
      if memberType = 'STUDENT' then fine := fine * 0.75;
      totalFines := totalFines + fine;
      writeln(memberName, ' fine: ', fine:0:2);
    end;
  writeln('Total fines: ', totalFines:0:2);
end.
```

Documentation order

- Cover sheet
- Table of contents
- Problem definition
- Algorithm
- Source code
- Trace table and test data
- Program execution screenshots
- Final review

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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